(133) 'Rethinking Greenberg's Universal 5 and the (imperfect) correlation between the nominal and clausal domain'. Dimitris Michelioudakis, Nikos Angelopoulos, Stergios Chatzikyriakidis, Natalia Chousou-Polydouri, Elena Anagnostopoulou. NELS 56 abstract (New York University).

Rethinking Greenberg's Universal 5 and the (imperfect) correlation between the nominal and clausal domain

1. Intro. This paper investigates Greenberg's Universal 5, an observation that has received virtually no theoretical or functional accounts, unlike other universals (Cinque 2005 on U20, Hawkins 1991,1992):

(1) **Greenberg's (1963) Universal 5:** If a language has dominant SOV order and the genitive follows the governing noun, then the adjective likewise follows the noun.

Greenberg's observation was only based on 50 languages. We conducted a more extensive study on a larger sample of typologically distinct languages. Our findings led us to the more general correlation in (2). This revised universal shifts the focus from head-directionality, instantiated by SOV orders in Greenberg's work, to subject placement (SV/VS). This, in fact, renders U5 a subcase of (2):

(2) **Generalized Universal 5:** VS significantly correlates with N-GEN. SV correlates with GEN-N unless nouns precede adjectives, in which case they correlate with N-GEN.

Our research also extends beyond word order to examine agreement patterns in genitive constructions featuring agreeing head marking. A consistent overlap was found between subject agreement in clauses and the agreement of the head noun with the genitive. This correlation leads us to propose that adnominal genitives structurally correspond to clausal subjects or, more precisely, the first Goal of Agree within the clause. We propose a formal account to integrate these agreement patterns with the word order facts in (2). This account bears on the current discussion on whether Agree is universally determined by a macroparameter (Baker 2008) or if similarities between clausal and nominal structures result from the clustering of functional head properties across different domains (see Roberts 2019).

2. Formalizing Generalized Universal 5. Beginning with the clausal domain, we adopt the standard assumption that a probe head exists above vP, e.g. T in Nom-Acc languages. This head probes and Agrees with the closest DP (the subject, S) and the features it copies are realized as agreement morphology on the lexical or the Aux-verb that this head attracts. It also attracts S above the inflected verb if it has an EPP-feature, giving rise to SV orders, or leaves S in a vP-internal position, as in some pro-drop languages, which gives rise to VS (Alexiadou & Anagnostopoulou 1998 i.a.). We propose that a probe also exists above nP that operates in the same manner; it probes the closest DP that serves as a possessor or adnominal argument and matches its phi-features. In languages with overt agreeing head-marking, the values it copies from the DP are overtly realized as an agreement affix on the head noun, such as the *t*-prefix on the head noun *íleve* 'flower' in (3) below from Garifuna.

(3) t- íleve n-u- fáluma *Garifuna* (adapted from Haurholm 2016: 65)
3F.SG.PREF1-flower 1SG.PREF1-EPENTH coconut 'the flower of my coconut'

The noun moves into the head, F in (4), which carries an EPP feature (+/-). Crucially, this EPP specification mirrors that of its clausal counterpart. +EPP attracts GEN to its Spec (GEN-N order, analogous to SV), (4a), while -EPP leaves GEN in a lower position (N-GEN order, analogous to VS), (4b).

(4) a. $[_{FP}$ GEN $[_{F'}$ F_{+EPP} +n $[_{nP}$ ~~GEN~~ $[_{n'}$ ~~n~~]]]] b. $[_{FP}$ F_{-EPP} +n $[_{nP}$ GEN $[_{n'}$ ~~n~~]]]

The analysis presented accounts for the correlation between verb-subject order and genitive-noun order observed in (2). The final observation in (2)–the possibility of GEN following N in SV languages–arises from N-raising. This movement is predicted in languages where N precedes the adjective.

3. Word Order. Analysis of 745 languages from Grambank confirms a statistically significant correlation between verb-subject and genitive-noun word orders. Specifically, 73.4% of VS languages display N-GEN order, and 77.5% of SV languages show GEN-N orders. Focusing on SV languages, 22.5% do not follow the proposed correlation, instead displaying N-GEN orders. This is attributed to N-raising, as nouns consistently precede adjectives in these languages. Adding Adj as a predictor for SV languages results in a remarkable increase of the correlation (from 77% to 93.9%, Cohen's d: 0.493).

4. Agreement Patterns. Besides word order, agreement patterns also suggest that genitives function as subjects. This study analyzed agreement in 153 head-marking languages (Inman et al., to appear; Chousou-Polydouri et al., 2023). These languages display total, partial or no overlap between subject agreement and agreeing head marking. Notably, however, agreeing head marking *never* coincides with object agreement. For instance, Garifuna is a case of total overlap (see table): it employs Prefix 1 for subject agreement, Prefix 2 for obliques, and a suffix for object agreement. The agreement marker *t-* for 3F.SG possessors (cf. (3) above) is identical to the corresponding S-agreement morpheme and the remaining poss-agreement markers precisely match the rest of the Prefix 1 paradigm. The table below summarises the alignment profiles of 38 N-GEN languages with total (20) or partial (18) overlap: the columns correspond to different alignments and the rows show which of S,A,P (S=intrans. subject, A=trans. subject, P=object) trigger agreement which morphologically overlaps with possessor agreement.

Series	1.SG	2.SG	3.M	3.F	1.PL	2.PL	3.PL
Simple forms							
Prefix 1	*n-*	*b-*	*l-*	*t-*	*wa-*	*hu-*	*ha-*
Prefix 2	*(A)na-*	*(A)bu-*	*(A)ni-*	*(A)nu-*	*(A)wa-*	*(A)nü:-*	*(A)nya-*
Suffix	*-na*	*-bu*	*-i*	*-u*	*-wa*	*-ü:*	*-nya*

	Nom-Acc	Erg-Abs	Tripartite	Active-Stative	Hierarchical
S and A	**20**	0	0	0	0
S and P	0	**3**	0	0	**9**
P only	0	0	0	2	**1**
A only	0	**2**	**1**	0	0

In Nom-Acc languages (**20**, column 2), poss-agreement aligns with subject agreement, exhibiting the same form as agreement morphology for A and S arguments. Also, possessor agreement patterns with what qualifies as the subject (i.e. S/P) in a significant proportion of Erg-Abs and Hierarchical languages: **3**/5 Erg-Abs (column 3) and **9**/10 Hierarchical languages (column 6). Following Bejar & Rezac (2009), we argue that in hierarchical languages, P is the first target of Agree. Equally straightforward is the overlap between poss-agreement and P-agreement in 2 active-stative (i.e. split-S) languages, as a subset of intrans. S's are indeed Ps. Less straightforward are: the overlap with P-agreement in Kutenai, a hierarchical language, A but not S agreement in **1** tripartite language, and with A-agreement only in Chol and Popti' (Mayan, but see Coon 2013, Deal & Royer 2025 for some relevant observations and potential account).

5. Macroparameters or microparameters. A central open question concerns the factors determining the structural configuration in which an agreeing head establishes a match with its target. One perspective posits a (on-off) macro-parameter that dictates whether Spec-head or downward (long-distance) Agree uniformly serves as the agreement configuration for both clausal and nominal heads, following Baker (2008). Alternatively, a uniform behaviour between clausal and nominal heads is simply preferred cross-linguistically as a result of acquisition strategies (e.g. Roberts' 2019 Input Generalisation) but not exceptionless. This intermediate state can be characterized as horizontally mesoparametric (Roberts *ibid.*). Distinguishing between these two approaches hinges on the quantity and nature of exceptions to the proposed generalization in (2). A macroparametric view would be reinforced if all the SV/N-GEN languages could receive principled explanations such as the systematic presence of N-raising. Similarly, evidence that all VS/GEN-N languages consistently feature finite V-raising over high projections, for instance, speaker-oriented adverbs, would strengthen this position. Conversely, a substantial number of unexplained exceptions would lend support to Roberts' mesoparametric account. Preliminary evidence suggests that all of these exceptions examined so far can receive such principled explanations thereby suggesting a macroparameter in Baker's sense.

Selected Literature. Baker, M. C. (2009). The macroparameter in a microparametric world. In *The limits of syntactic variation*. **Cinque, G. (2005).** Deriving Greenberg's Universal 20 and its exceptions. LI. **Coon, J. (2013).** *Aspects of split ergativity*. Oxford University Press. **Deal, A. R., & Royer, J. (2025).** Mayan animacy hierarchy effects and the dynamics of Agree. NLLT. **Roberts, I. G. (2019).** *Parameter hierarchies and universal grammar*. Oxford University Press.